

# When Can Language Models Join Without Relation Names?

## Unique-Path Traversal versus End-to-End Accuracy under Path Ambiguity\*

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### Abstract

We ask whether 2-hop answering over small labeled graphs needs readable relation names, or only equality among identifiers. Opaque isomorphic reasoning (OIR) replaces relation strings with keyed opaque tokens, withholds a legend, and scores the answer node rather than an emitted  $(r_1, r_2)$ . HMAC-SHA256 is one generator of that renaming on each instance’s finite atom set, not a cryptographic or privacy primitive. Unique sealed routes still join (OpenAI unique English/opaque 200/200; hashed-id no-ARM 200/200 at 512 tokens; opaque unique 32/32 with the start in  $q$ ; both-opaque unique 32/32 on three families). Under two competing sealed routes, a prompt that names the condition and uses readable case ids (named-arm) usually yields UNKNOWN, not the decoy: OpenAI 38/200 gold, UNKNOWN 151, decoy 11. The same graphs with hashed case ids (ARM line kept) are 134/200 (UNKNOWN 44, decoy 20). Two-path exact-match cannot be attributed to a join over CONTEXT. Dropping the condition header, at 4096 tokens and reasoning effort medium, cyclic-decoy English two-path is 197/200 and opaque two-path is 174/200. Unique no-ARM at 512 is not a matched-budget contrast with two-path no-ARM at 4096/medium. English relation names restore OpenAI with anonymous entities (opaque-entity two-path 32/32 vs. both-opaque 1/32). On the primary  $2 \times 2$ , seals hide relation strings; entity names remain. They are not a confidentiality proof.

## 1 Introduction

The scored Wikidata question is *What city is the headquarters of the company led by Michael Eisner?* English unique-path context walks **works\_at** then **headquartered\_in** to Burbank. That success may use English predicates, or only equality among identifiers. The two-path decoy is an injected second person→company→city route (**partner\_of/located\_in**) to Kansas City, not a Wikidata fact. Opaque two-path keeps the English question and entity names; those predicates become HMAC tokens (**works\_at**  $\mapsto$  E334675310f05 on the scored Eisner item).

**Opaque isomorphic reasoning (OIR)** measures whether a model can join after English predicate strings are replaced by keyed opaque tokens, with no legend in the prompt. A 2-hop factors as two inferred maps; we score only the answer node: selection  $q \mapsto (r_1, r_2)$  and traversal  $(v_0, r_1, r_2, G) \mapsto v_2$ . Opacity does not, by itself, block traversal. The question is when the prompt no longer maps query semantics onto one of two sealed relation sequences. Putting the start entity in  $q$  as the same atom as in  $V$  (**start in  $q$** ) separates that question from start-entity identification. On the primary  $2 \times 2$ , “lexical cues” means relation strings (entities stay English); both-opaque is a separate ablation. That suite is WIKI-H5. **Named-arm** means a one-line condition header plus readable case ids (**F22\_OPAQUE\_AMBIG\_** $i$ ); hashed-id cells replace those ids with hashes. A **lock** is a scored JSON we do not overwrite; a **freeze** is a fixed record list.

\*Large language models were used as writing assistants for grammar and phrasing. All ideas, claims, experimental design, locked numbers, and scientific conclusions are the authors’.

Figure 1 plots the  $n=32$  named-arm lock beside the city-typed  $n=200$  freeze. This paper reports three measurements. (i) Unique sealed routes still join with the start in  $q$  and no written hop. (ii) Under two competing sealed routes the usual *named-arm* reply is UNKNOWN, not the decoy; exact-match still moves with prompt surface, including a no-ARM hashed-id cyclic English/opaque pair at 4096/medium (Section 5.2). (iii) Matched demonstrations restore shown seals; novel seals are a negative control whose expected gold is near zero. What is new is how opacity meets path ambiguity under a named-arm prompt (Section 5.1), and how large a prompt-surface move can be on the same sealed graph (Section 5.2).

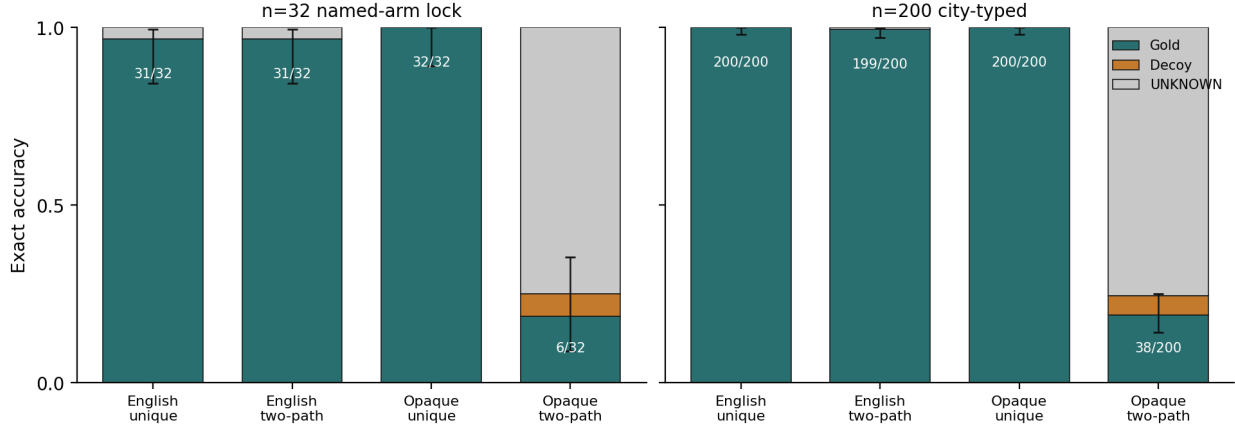


Figure 1: OpenAI WIKI-H5 2×2 (start in  $q$ ). Stacked bars are gold / decoy / UNKNOWN; whiskers are Wilson 95% intervals. Left:  $n=32$  named-arm lock (August; opaque two-path 6/32). Right: city-typed  $n=200$  freeze (named-arm opaque two-path 38/200, UNKNOWN 151, decoy 11). Hashed-id and alias  $k=3$  are Table 3 and Sections 5.1–5.2.

## 2 Related Work

Think-on-Graph and Reasoning-on-Graphs select the next readable relation (Sun et al., 2024; Luo et al., 2024). Ciphered ICL is not path ambiguity on a labeled graph (Fang et al., 2025; Guo et al., 2026). GSM-Symbolic swaps entity names in English math and templates some numerical values (Mirzadeh et al., 2025); 2WikiMultihopQA is a readable multi-hop control (Ho et al., 2020).

ARoG anonymizes *entities* while relations stay the retrieval language (Ning et al., 2026). PrivGemo HMAC-seals entities ( $\phi_Q$ , e.g. `ent_17`); relations stay original labels or clustered schema words, not keyed opaque tokens, and a local Hand answers on the raw graph after a remote Brain ranks retrieved paths (CWQ 67% plaintext  $\rightarrow$  59% fully anonymized; WebQSP 75% fully anonymized) (Tan et al., 2026; Talmor and Berant, 2018; Yih et al., 2016). CoR explores readable relations because entities often have no semantics beyond opaque IDs — CoR’s opaque-entity hide-set, not relation-only WIKI-H5 (Liu et al., 2026). Tang et al. (2023) find LLMs fail when relation names are replaced by symbols (`r1`) and rules must carry deduction; OpenAI unique-path opaque 32/32 is the complementary case: sealed relations still traverse when topology fixes the route. Guo et al. (2025, section 2.2) report collapse to random guessing among end entities when distractor two-hops have other sources. Named-arm Wiki-H5 opaque two-path, with UNKNOWN allowed, is abstention rather than guessing (Section 5.1, Table 3). A forced-choice arm is not in this evaluation (Section 7) and would put those protocols on the same scale. OIR withholds a retrieved-path ranking and a local plaintext decoder on the free arms (Table 1); both 2-hops still appear in CONTEXT.

Chain-of-thought and Text-to-SQL externalize plans over English (Wei et al., 2022; Yu et al., 2018). We ask a behavioral question: can a pretrained model join when the relation vocabulary is freshly randomized and no plan is given? A sealed plan is then an execution upper bound. NLGraph and Talk like a Graph encode graphs without readable typed predicates (Wang et al., 2023; Fatemi et al., 2024); OIR starts from a labeled KG and removes only the relation lexicon. Case-id strings that name the condition (OPAQUE\_AMBIG) are semantic content, not only a meaning-preserving format perturbation in the sense of Sclar et al. (2024); we

Table 1: Related protocols. PrivGemo HMAC-seals entities; relations remain original or clustered schema labels, and a local Hand answers on retrieved paths (CWQ/WebQSP). OIR withholds that ranking and local decode on the free arms and scores the answer token from the same model; named-arm two-path exact-match is not a CWQ rate. Both 2-hops still appear in CONTEXT.

| Work       | Hidden                                                                                   | Remote task                                            | Join under opacity         |
|------------|------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------|
| ICL        | substitution text                                                                        | ICL task accuracy                                      | ICL tasks                  |
| CIPHERS    |                                                                                          | (bijection inferred)                                   |                            |
| ARoG       | entities; relations<br>readable; topic<br>names in $q$                                   | retrieve + generate<br>(relations,<br>concepts)        | privacy KG-RAG             |
| PrivGemo   | entities (HMAC);<br>relations original<br>or clustered<br>schema words                   | rank retrieved<br>paths; local<br>plaintext answer     | privacy KG-RAG             |
| CoR        | opaque entity<br>IDs; relations<br>readable                                              | relation-centric<br>exploration                        | KGQA                       |
| <b>OIR</b> | relations<br>(primary $2 \times 2$ );<br>entities too<br>(ablation); one<br>join or none | emit<br>graph-namespace<br>answer (no local<br>decode) | selection vs.<br>traversal |

still treat packing as a factor. Symbol tuning, binding-ID analyses, and compositional-limit studies change the labels or the task (Wei et al., 2023; Feng and Steinhardt, 2024; Dziri et al., 2023). OIR holds the join, hop length, triple count, and question template fixed and replaces relation strings; OpenAI unique-path opaque traversal stays at 32/32, so that removal withdraws the cue that would otherwise select between routes rather than blocking execution.

### 3 Problem Formulation

Let  $G = (V, R, T)$  be a labeled multi-relational graph ( $T \subseteq V \times R \times V$ ) and  $q$  a natural-language or programmatic query with answer  $a \in V$ . A **seal**  $\sigma$  is a keyed map from entity and relation strings to opaque tokens that we verify to be injective on the finite normalized atom set of each evaluation instance. We reject any instance where two distinct raw atoms normalize to the same string. Truncated HMAC-SHA256 (Krawczyk et al., 1997) is one generator of  $\sigma$ , not a cryptographic bound (Appendix A). The model receives  $G$  or  $\sigma(G)$  and a query form built from  $\sigma$  (or from English, depending on the arm) and must return the answer node in that namespace; recovering  $\sigma^{-1}$  is not scored.

A 2-hop join factors as two maps we measure separately:

$$\text{selection: } (q, G, \mathcal{C}) \mapsto (r_1, r_2), \quad (1)$$

$$\text{traversal: } (v_0, r_1, r_2, G) \mapsto v_2. \quad (2)$$

Traversal is set-valued ( $\tau_G: V \times R \times R \rightarrow 2^V$ ). We evaluate only instances for which  $|\tau_G(v_0, r_1, r_2)| = 1$  for the gold plan (and, on two-path items, for the decoy plan), so each scored route has a unique tail. Uniqueness is per plan, not “one answer in  $G$ .” A sealed join plan supplies  $(r_1, r_2)$  and tests (2). The CEO free-English arm hashes  $q$  piecewise, so  $v_0$  typically does not appear as a single graph atom; it is not a pure test of (1).

A 2-hop is *structurally executable* when  $(r_1, r_2)$  is given, or when topology leaves only one 2-hop from a start already in  $q$ . The prompt *explicitly maps* query semantics onto one relation sequence when it contains English predicates, a legend, or demonstrated seals. When two complete routes exist and relation tokens are arbitrary HMAC strings with no such mapping, selection is *unsupported by explicit cues* under this construction. Residual statistical structure can remain (token form, order, HMAC vs. permutation rates);

we do not claim mathematical unidentifiability. Any gold on that arm must use residual prompt cues, parametric memory of the named entities, or other unmeasured information; we do not observe  $(r_1, r_2)$ . Low exact-match is not evidence that traversal failed. Dual-path separates the cases: shown seals restore end-to-end accuracy; novel seals are a negative control (expected gold near zero). The scored output is the answer node or UNKNOWN, not  $(r_1, r_2)$ . We infer whether selection was supported from those outcomes and from controls (unique topology, written hop, matched vs. novel seals). Throughout, “selection failure” names that inference about the latent  $(r_1, r_2)$  stage; we do not observe the pair. Models may abstain (UNKNOWN) rather than guess (selective prediction; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). Coverage is the fraction of items with a non-UNKNOWN parse (other and both-named count; empty completions stored as UNKNOWN do not; NO\_OUTPUT does not); selective accuracy is exact match among those items, not a forced-choice among the two routes (El-Yaniv and Wiener, 2010; Wen et al., 2025). NO\_OUTPUT is an empty completion on a runner that does not store empties as UNKNOWN (Wiki-H5/Wiki-OO). **Both-named** is a parse that emits both person tails.

After sealing, the Eisner unique-path CONTEXT is `Michael_Eisner | E334675310f05 | The_Walt_Disney_Company` then `The_Walt_Disney_Company | Eac04fab538ca | Burbank` (12-hex tokens; entities remain English on the primary  $2 \times 2$ ).

We use five measurement arms. **Readable English** is the baseline (start in  $q$ ; names and predicates unsealed). **Unique opaque** has one 2-hop and no plan: execution when topology uniquely determines the route. **Two-path opaque** has two same-type 2-hops and no plan: end-to-end behavior when two complete routes compete and the prompt supplies no semantic-to-symbol map. **Sealed join plan** writes  $(r_1, r_2)$  over  $\sigma$ -atoms (execution control); **SealRouter** is the non-LLM exact joiner on that sequence. **Free sealed English** seals the question and graph with no plan and no legend. On CEO, the question is hashed piecewise, so the full person atom in  $G$  typically does not appear in  $q$  (confounded; not the matched factorial). Additional scaffolds (name/key legends, sealed demonstrations) appear as ablations.

## 4 Method

**Protocol terms.** Headline WIKI-H5 quizzes are **isolation** files (one item per file; gold stored outside the prompt). Batch  $n=12/n=24$  three-arm files use one shared key (Appendix A). A **lock** is a scored JSON we do not overwrite without `--force`. A **freeze** is a fixed record list. An **ARM** line is a one-line header that names the condition. **Named-arm** packing uses readable case ids plus that header; **hashed-id** packing keeps the ARM line and hashes the case ids; **no-ARM** packing drops the header. Unless noted, OpenAI uses 512 completion tokens with `reasoning_effort` omitted; **4096/medium** is 4096 completion tokens and `reasoning_effort` medium. Alias  $k=3$  is three draws of the unpinned snapshot; the English-entity ambiguity sweep  $k=1.5$  is a different  $k$  (Appendix B). Run tags and lock filenames are Appendix A. WIKI-H5 is the relation-only  $2 \times 2$ ; header H5 keeps that ARM line.

**Seal recipe.** Atoms are normalized then HMAC-SHA256-renamed with a per-item 16-byte key (token E plus 12 hex digits). HMAC is keyed renaming, not a security mechanism; 48-bit truncation is instance-wise injective on these small atom sets (normalization rules, type non-separation, and collision checks: Appendix A). The intervention removes lexical semantics and also changes token-level surface statistics; permutation and alphanumeric generators assess that latter variation (Table 8). Hashing  $q$  token-by-token does not reconstruct the graph atom `Michael_Eisner`; that is the missing-start failure. Reported opaque cells are this HMAC construction, not every opaque identifier distribution.

**Matched  $2 \times 2$  (start in  $q$ ).** The  $n=32$  suite is a seeded random sample of 32 of 73 Wikidata-derived CEO records (`random.Random(20260821).sample`; Vrandečić and Krötzsch, 2014), not a published CEO benchmark and not a hand-picked subset. Each instance is 4 triples (unique-path: 6 entities with a disconnected decoy island; two-path: 5 entities sharing the start). The gold two-hop uses `works_at` then the headquarters relation; the question says “led by”. The two-path condition *injects* a competing route: `ambig_edges` attaches record  $i+1$ ’s company/HQ under `partner_of/located_in`. That decoy is not a Wikidata fact. It isolates same-type competition on a fixed template, not naturally occurring CWQ/WebQSP ambiguity. To isolate relation selection from start-entity identification we keep the start string in the question and hold hop length,

template, and triple count fixed (4 triples; unique-path has 6 entities, two-path 5). The intended  $2 \times 2$  factors are relation labels (English vs. HMAC) and topology (one 2-hop from the start vs. two). Named-arm cells also change the ARM line and a readable case id with the condition (lines below); hashed-id and no-ARM packing are Section 5.2. Tails are intended as the same type; Wikidata P159 (headquarters location) was not filtered (buildings, stations, a street, and one state among gold HQs). Both OpenAI opaque two-path decoys sit on non-city golds with a city decoy tail (Appendix A). On Wikidata two-path graphs both intermediates are companies (`works_at/headquartered_in` vs. `partner_of/located_in`). In the primary run the gold 2-hop is listed before the decoy 2-hop in CONTEXT (same order in English and opaque). Listing-order robustness is Section 5.2: a new draw on the  $n=200$  freeze, not a reordering of the primary completions. The  $n=32$  shuffle is a later snapshot, not that comparison (Appendix A).

The primary two-path prompt is a single user message:

```
MODEL UNDER TEST. Read ONLY this file. Use CONTEXT only. No other files.
Format: ANSWER_PLAIN[...]: <city_or_UNKNOWN>
ARM OPAQUE_ambiguous: opaque relations, English entities;
two 2-hops from start.
```

Named-arm ARM lines: ENG\_UNIQUE one 2-hop; ENG\_ambiguous two 2-hops; OPAQUE\_UNIQUE opaque one 2-hop; OPAQUE\_ambiguous two 2-hops (quoted); OPAQUE\_ambiguous\_PLAN same graph plus the sealed sequence. OE/OO Format is ANSWER\_SEALED; ARM lines, verbatim from the isolation quizzes (same 32 graphs; start seal in  $q$ ):

```
ARM OE_UNIQUE: opaque entities, English relations; one 2-hop.
ARM OE_ambiguous: opaque entities, English relations; two 2-hops.
ARM OO_UNIQUE: opaque entities and relations; one 2-hop; start seal in q.
ARM OO_ambiguous: opaque entities and relations; two 2-hops; start seal in q.
```

$n=200$  named-arm ids are F200\_⟨arm⟩\_i;  $n=200$  hashed-id ids are H200\_ plus 10 hex;  $n=32$  hashed-id ids use prefix HDA\_ (Section 5.2). That primary prompt does *not* contain the equality-recipe line (Appendix J). The format token is <city\_or\_UNKNOWN>, but six  $n=32$  gold HQs are not cities (Appendix A). A model that obeys the format on those items must emit UNKNOWN. The powered city-typed cell is  $n=200$  (Section 5.1); the  $n=32$  lock is the three-family pin. OpenAI English two-path is 31/32 under that CONTEXT-only instruction, so the line does not by itself force abstention. Readable names (Eisner, Burbank) still leave OpenAI at UNKNOWN 24/32 on opaque two-path: either parametric HQ is treated as inadmissible, or the verb does not map onto one seal. We cannot separate those readings here (Section 7). The CEO three-arm decoy 2-hop begins at a different person; it is not the matched two-path factorial.

**Further constructions.** The relation-only factorial keeps entities English. A second isolation suite on the same 32 CEO records seals entities, relations, or both, always with the start token in  $q$ . Opaque-entity / English-relation (OE) is the ARoG-like cell: readable predicates, anonymous nodes. Both-opaque (WIKI-OO) seals entities *and* relations; it is a complementary ablation, not the primary relation-only  $2 \times 2$ . Readable entity names therefore leave residual pretrained world knowledge as a possible cue; both-opaque is not a substitute for synthetic identifiers.

WikiMovies facts are sampled from the public MetaQA movie KB (Miller et al., 2016; Zhang et al., 2018). MOVIE-A32 and MOVIE-A100 leave English verbs such as *directed* in  $q$ ; MOVIE-B hashes question atoms except the bracketed start entity. We exclude a genre-vs-language variant because object type would bypass relation selection. This is not an evaluation on the official MetaQA 14K test split. Dual-path holds  $G$  fixed with two complete 2-hop routes and varies whether demonstrations expose the gold seals (Appendix D).

**Scoring.** The format asks for one answer token in the graph’s namespace or UNKNOWN (plaintext city when entities stay English;  $\sigma(a)$  when entities are sealed). Primary metric: exact match (UNKNOWN is a miss). Scored UNKNOWN on Wiki-H5/Wiki-OO is an explicit UNKNOWN line. Dual and some no-hop OpenAI cells stored empty completions as UNKNOWN. On two-path arms we also report decoy and UNKNOWN rates (Table 3). Wilson 95% intervals use  $z=1.96$  (Wilson, 1927) ( $0/32 \Rightarrow [0.00, 0.11]$ ;  $6/32 \Rightarrow [0.09, 0.35]$ ;  $32/32 \Rightarrow [0.89, 1.00]$ ). Those

intervals treat the 32 items as Bernoulli trials on a single draw; they are not a generation-variance interval. OpenAI WIKI-H5  $k=3$  at  $n=32$  and  $n=200$  is Section 5.1. McNemar  $p$ -values below are unadjusted across the reported contrasts. On the OpenAI Wikidata 2×2, English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar exact two-sided complete-case  $p=5.96\times 10^{-8}$ ; McNemar, 1947). That is the two-path paired contrast, not an interaction test, and it largely measures removing the cue that made one sealed route identifiable. The empirical content is the response mode (UNKNOWN vs. decoy) and prompt-surface packing (Section 5.2). Isolation  $n=32$  uses one quiz per file and a per-item HMAC. Cells marked **n.r.** were not evaluated; we never impute them as zeros.

**Models.** The primary family is OpenAI **gpt-5.6-so1** (Chat Completions; unpinned alias; temperature omitted; 512 completion tokens unless noted). Composer 2.5 and Grok 4.5 used Cursor’s agent SDK on the same quiz files, with a reply-only preamble OpenAI never received. Unique-path opaque still executes on those SDK cells (Composer 27/32, OpenAI 32/32, Grok 30/32; Table 7). Tools, retry, and empty-reply handling are not equated. The August lock stores the parsed answer, not the API `model` string. Inference settings and  $k=3$  alias draws are Appendix A. A packaging-only Claude probe is Appendix F. Isolation quizzes and locked scores are at the supplementary archive.

## 5 Results

Table 2 names the cells. OpenAI is primary; Composer/Grok used a different harness (Table 7). MOVIE-B is a cross-dataset check, not a Wiki-H5 resample. Packaging probes, plus Spider  $n=100$  and 2Wiki-CF  $n=200$ , are Appendix G. JSON no-hop binding cells are Appendix E.

Table 2: Cell inventory. WIKI-H5-200 is the powered city-typed freeze (OpenAI). Composer/Grok  $n=32$  SDK cells: Table 7. Six  $n=32$  P159 golds are not cities (Appendix A). A100 Condition A uses a thinner ARM than A32. Isolation cells use per-item HMAC.

| ID               | Dataset                              | Condition                       | $n$     | Models               | Scored      |
|------------------|--------------------------------------|---------------------------------|---------|----------------------|-------------|
| WIKI-H5          | Wikidata CEO                         | rel-only 2×2, start in $q$      | 32      | OpenAI (primary)     | node        |
| WIKI-H5-200      | Wikidata CEO, city P31 (instance of) | same 2×2, QID freeze            | 200     | OpenAI               | node        |
| Header×decoy     | same 32 people                       | ARM line × decoy, hashed ids    | 32      | 3 families           | node        |
| Header×decoy-200 | same 200 people                      | ARM × decoy, hashed ids         | 200     | OpenAI               | node        |
| Unique-no-ARM    | same people                          | unique path, hashed ids, no ARM | 32, 200 | OpenAI               | node        |
| WIKI-OO          | same graphs                          | both-opaque                     | 32      | 3 families           | $\sigma(a)$ |
| WIKI-OO-200      | $n=200$ freeze                       | both-opaque                     | 200     | OpenAI               | $\sigma(a)$ |
| WIKI-SHUF        | same two-path                        | listing order                   | 32      | OpenAI               | city        |
| WIKI-SHUF-200    | $n=200$ freeze                       | listing order                   | 200     | OpenAI               | city        |
| MOVIE-A32        | WikiMovies                           | actor→movie→dir vs writer       | 32      | 3 models             | person      |
| MOVIE-A100       | WikiMovies                           | same hops; verbs in $q$         | 100     | OpenAI; Comp./Grok A | person      |
| MOVIE-A200       | WikiMovies                           | $n=100$ prefix + 100 new actors | 200     | OpenAI               | person      |
| MOVIE-B          | WikiMovies                           | hashed verbs + two-hop ARM      | 100     | OpenAI               | person      |
| MOVIE-B200       | WikiMovies                           | Condition B on $n=200$ freeze   | 200     | OpenAI               | person      |
| DUAL             | WikiMovies/synth                     | matched vs. novel demos         | 32      | 3 models             | $\sigma(a)$ |
| DUAL-200         | WikiMovies $n=200$                   | matched vs. novel               | 200     | OpenAI               | $\sigma(a)$ |
| CEO-NL           | Wikidata CEO                         | piecewise $q$ , no plan         | 32      | 3 models             | $\sigma(a)$ |
| CEO-NL-200       | $n=200$ freeze                       | piecewise $q$ , no plan         | 200     | OpenAI               | $\sigma(a)$ |

### 5.1 Opacity × path ambiguity

WIKI-H5 holds the start string in  $q$ . The intended factors are relation lexicality (English vs. HMAC) and the number of 2-hops from that start (one vs. two). Named-arm packing (ARM line and readable case ids) also changes with the condition; hashed-id and no-ARM cells are Section 5.2. OpenAI is the primary family. Composer 2.5 and Grok 4.5 used Cursor’s agent SDK (Table 7).

**City-typed  $n=200$  (OpenAI).** The  $n=32$  source file has 73 CEO rows and six non-city P159 golds. WIKI-H5-200 is a new freeze: 200 unique people and companies, HQ frequency capped at 3, every HQ an instance of a city-like Wikidata class (P31; list in Appendix A), person/company/HQ QIDs stored, and all 200 P169 (chief executive officer)+P159+P31 triples re-checked live against the Query Service. It does not overwrite

the  $n=32$  lock. OpenAI named-arm counts are Table 3 (opaque two-path Wilson [0.14, 0.25]); plan 200/200. Hashed-id unique-path with no ARM (same HMAC CONTEXT as OPAQUE\_UNIQUE; 512 tokens) is 200/200: the unique-route join does not need the ARM that names one 2-hop. That named-arm two-path rate sits inside the  $n=32$  Wilson band for 6/32. Decoy 0 on the  $n=26$  city slice was a small-sample artifact:  $n=26$  cannot resolve a  $\sim 5\%$  decoy rate. At  $n=200$  the usual named-arm reply is still UNKNOWN ( $\approx 14:1$  vs. decoy). This is not a CWQ/WebQSP number: those protocols retrieve paths and answer locally; OIR withholds a retrieved-path ranking and a local decoder on the free arms (Table 1). Prompt-surface packing of this freeze is Section 5.2.  $n=200$  is OpenAI-only. Headline locks and  $k=3$  draws are Table 3 and Appendix A. Three-family cells stay at  $n=32$ .

**$n=32$  named-arm lock.** On unique sealed routes OpenAI executes (opaque unique 32/32; plan 32/32). Hashed-id unique-path with no ARM is 27/32 (26/26 city golds); the five misses are non-city P159 format tokens that named-arm unique emitted. On competing sealed routes the usual named-arm reply is UNKNOWN, not the decoy. On 26 city-valued items unique English, unique opaque, and English two-path are 26/26; named-arm opaque two-path is 5/26, 2/26, 2/26 across three draws. August: UNKNOWN 21, decoy 0, selective accuracy 5/5. The lock is opaque two-path 6/32 (UNKNOWN 24, decoy 2); both decoys sit on non-city P159 golds. Wilson:  $6/32 \Rightarrow [0.09, 0.35]$ ; city slice  $5/26 \Rightarrow [0.09, 0.38]$ . The left panel of Figure 1 is this  $n=32$  lock.

Let  $\Delta_{\text{opq}}$  be opaque minus English exact-match on  $n=32$ : unique +1/32, two-path -25/32 (25 English-correct and opaque-wrong, 0 reverse; McNemar  $p=5.96 \times 10^{-8}$ ). Opacity is not a randomized within-item treatment (Appendix I).

**OpenAI  $k=3$ .** Alias draws of the unpinned snapshot are not a generation-variance interval. Each packing’s  $k=3$  is that packing’s headline lock plus two later alias draws of the same packing. At  $n=32$ : August named-arm lock 6/32 (Composer 10/32, Grok 1/32); 15 September alias draws 3/32, 3/32 (Composer 7, 6; Grok 0, 0). Named-arm  $k=3$  is that August headline plus the two 15 September alias draws (6/3/3); the 11 September named-arm 5/32 is the same-date hashed-id pair, not a later alias of the August headline, so it is not in that triple. Hashed-id  $k=3$  is 20/21/22 because the hashed-id headline is the 11 September lock 20/32, plus 15 September 21, 22 (512 tokens, `reasoning_effort` omitted). OpenAI named-arm draws 2–3 are both 3/32 and Grok both 0/32; Composer is 7 then 6. August named-arm is highest in all three families. At  $n=200$  the headline pair and  $k=3$  golds are Table 3. Those later draws do not replace the headline locks (Appendix A). On OpenAI the other  $n=32$  named-arm cells hold: unique opaque 32/32, 31/32, 32/32; English two-path 31/32, 30/32, 30/32; plan 32/32 on every draw. Named-arm English two-path is OpenAI 31/32, Composer 26/32, Grok 4/32: a packaging $\times$ family effect, not a ranking. Opaque unique stays high (Composer 27/32, Grok 30/32); a written hop recovers (28/32, 32/32). Two-path coverage / selective accuracy (named-arm): OpenAI opaque 8/32, 6/8; Composer 10/32, 10/10; Grok 1/32, 1/1. Two other generators still have unique  $>$  two-path (Table 8;  $n=32$  header $\times$ decoy is Table 9).

Table 3: OpenAI response modes. Named-arm uses readable case ids. Hashed-id 512 keeps the ARM line (remainder: other 1, NO\_OUTPUT 1). Coverage  $49/200 \rightarrow 155/200$ ; selective accuracy  $38/49 \rightarrow 134/155$ . Alias  $k=3$  gold 38/46/43 vs. 134/130/127 (gaps 96, 84, 84); coverage  $49/56/58$  vs.  $155/153/145$ . Same-date  $n=32$  hashed-id is 20/1/8 (other 2, NO\_OUTPUT 1) at 512 tokens; Table 9’s H5-cyclic 20/32 is 4096/medium, a different run. Gold on all three  $k=3$  draws is named-arm 28, hashed-id 97. Composer/Grok SDK cells: Table 7.

| OpenAI Wiki-H5 (gold / decoy / UNK) | Named $n=32$ | Named $n=200$ | Hashed $n=200$ |
|-------------------------------------|--------------|---------------|----------------|
| English unique                      | 31/0/1       | 200/0/0       | —              |
| English two-path                    | 31/0/1       | 199/0/1       | —              |
| Opaque unique                       | 32/0/0       | 200/0/0       | —              |
| Opaque two-path                     | 6/2/24       | 38/11/151     | 134/20/44      |

## 5.2 Prompt surface as a factor

The named-arm WIKI-H5 prompt contains this line:

ARM OPAQUE\_AMBIG: opaque relations, English entities; two 2-hops from start.

That sentence tells the model in English that two 2-hops exist. It is an uncontrolled treatment if abstention is attributed only to opacity-meets-ambiguity. Two manipulations—ARM line and case-id wording—must not be collapsed.

**Case-id wording, ARM line held fixed.** The case id appears in `Format: ANSWER_PLAIN[id]` and in the ID line. Named-arm uses readable ids (`F22_OPAQUE_AMBIG_i`); the H5-cyclic arm substitutes hashes. ARM line and `CONTEXT` are identical. On the city-typed  $n=200$  freeze, the same 512-token contrast is Table 3 (+96 golds; hashed-id has one `NO_OUTPUT`). Coverage and selective accuracy in that table show the gain is not only more guessing. Decoy also rises  $11 \rightarrow 20$ . Headline  $n=200$  is one lock each (Appendix A). Alias  $k=3$  at  $n=200$  (Table 3) does not make the cells meet. The  $n=32$  pair is the named-arm and hashed-id triples in Section 5.1. The  $n=32$  same-date pair is 11 September (512 tokens, `reasoning_effort` omitted): 5/32 vs. 20/32 (+15; one `NO_OUTPUT` each). Both prompts keep the ARM line, so this isolates case-id wording, not whether the prompt announces two 2-hops: hashing the id removes `OPAQUE_AMBIG` from `Format/ID`; the ARM line remains. Item 0 named-arm `Format` is `ANSWER_PLAIN[F22_OPAQUE_AMBIG_0]`; hashed-id H5 is `ANSWER_PLAIN[HDA_4eba493d71]`. Against the named-arm triple only +18 and +19 are same-day.

**ARM line, case ids held hashed.** Cyclic decoys attach record  $i+1$ ’s company/HQ, as in WIKI-H5. Constant decoys reuse the first pool record that does not share company, HQ, or person with the gold (`curve_decoy`). On  $n=32$  that record is Cogent Communications  $\rightarrow$  Washington, DC on every item (gold=decoy is 0). On the  $n=200$  freeze it is San\_Jose on 199/200 items and London on 1 (item 0, gold San\_Jose); gold=decoy is 0; Cogent is not that decoy. Cyclic gold=decoy is 0 at both  $n=32$  and  $n=200$ . Table 4 crosses the ARM line with cyclic vs. constant decoys at 4096/medium on the  $n=200$  freeze. H5 cyclic vs. constant McNemar 95 vs. 10,  $p=1.6 \times 10^{-18}$ ; no-ARM 30 vs. 7,  $p=1.9 \times 10^{-4}$ . The  $n=32$  header $\times$ decoy cell (Table 9) already showed a simple-effect gap under H5 (cyclic 20 vs. constant 10; McNemar 14 vs. 4,  $p=0.031$ ) and not under no-ARM (gold 25 vs. 27; McNemar 1 vs. 3,  $p=0.625$ ). At  $n=200$  both H5 and no-ARM cyclic-vs-constant simple effects are significant; we do not report an item-level interaction test. Of the 95 items that are gold under H5-cyclic and not under H5-constant, 74 become UNKNOWN and 21 take the constant decoy (McNemar 95 vs. 10). At  $n=32$ , all 14 cyclic-only golds become UNKNOWN (none take Cogent $\rightarrow$ DC). The drop is mostly abstention, not selection of the constant decoy’s HQ. Isolation files show one item, so the model cannot see that the decoy repeats across the freeze. We still do not identify a mechanism. K2 is a shorter ARM that still names two hops:

ARM K2: opaque relations; 2 same-type 2-hop(s) from start.

K2 cyclic is near no-ARM cyclic (Table 4), so announcing two 2-hops is not what produces the H5 drop. `ENG_AMBIG` and `OE_AMBIG` carry that two-hop label and still score at ceiling (OpenAI 31/32 and 32/32;  $n=200$  English two-path 199/200). Which H5 tokens matter (condition label vs. “English entities”) is not identified here. Two-path K2 and no-ARM were not run at 512.

**Listing order.** Within one shuffle run on the  $n=200$  freeze, gold-first is 25/100 and decoy-first is 34/100 (two-proportion  $p \approx 0.16$ , not significant). That is the listing test: same draw, same case-id family. Overall 59/200 versus the primary named-arm cell (Table 3) is a new draw (case ids `F200S_` rather than `F200_`), not a within-run reordering effect; we do not read +21 as a listing-order gain. The  $n=32$  shuffle is a later snapshot (Appendix A): opaque two-path 5/32 (gold-first 3/16, decoy-first 2/16); decoy 0/16 vs. 1/16.

**English Grok.** Named-arm English two-path is Grok 4/32 (UNKNOWN 28). The English header $\times$ decoy cell (`ENG_NONE_CYC`) is 24/32. That cell has *no* ARM line and hashed ids, so two things differ from the 4/32 named-arm English prompt. Call it English, no ARM, hashed ids—not “arm-neutral English.” A



Table 4: Header×decoy on the city-typed  $n=200$  freeze (same people as WIKI-H5-200). Cyclic = record  $i+1$ ’s HQ; constant = `curve_decoy` ( $n=200$ : San\_Jose  $\times 199$ , London  $\times 1$ ; not Cogent→DC). English, no ARM, cyclic is 197/200. No-ARM cyclic English vs. opaque is McNemar 25 vs. 2 ( $p=5.65\times 10^{-6}$ ). H5 cyclic vs. constant McNemar 95 vs. 10 ( $p=1.6\times 10^{-18}$ ). H5 vs. no-ARM on constant decoys is McNemar 4 vs. 111. H5 vs. no-ARM on cyclic decoys is McNemar 6 vs. 51 ( $p=5.7\times 10^{-10}$ ). No-ARM cyclic opaque is gold 174, decoy 17, UNKNOWN 5, other 4. The 512-token hashed-id H5-cyclic lock is 134/200 (one NO\_OUTPUT). Cyclic gold=decoy is 0 at  $n=32$  and  $n=200$ .  $n=32$  golds: Table 9.

| ARM line (OpenAI, hashed ids, 4096/medium, $n=200$ ) | Cyclic  | Constant |
|------------------------------------------------------|---------|----------|
| no ARM                                               | 174/200 | 151/200  |
| H5                                                   | 129/200 | 44/200   |
| K2                                                   | 179/200 | 144/200  |

prompt-verbatim Grok rerun of the named-arm English files stays 4/32, so the Cursor preamble is not that gap.

What the data support: exact-match under competing sealed routes is strongly sensitive to prompt surface form. Case-id wording and the ARM line are large moves whose shared cause, if any, is not identified here (paragraphs above; Table 4). Budgets differ, so these are not a ranking of manipulations.

A consequence for the contrast with Guo et al. (2025, section 2.2). That work reports collapse to guessing among end entities when distractor two-hops have other sources. Our named-arm OpenAI cell is abstention (UNKNOWN 24/32, decoy 2/32 at  $n=32$ ; UNKNOWN 151, decoy 11 at  $n=200$ ), not a coin-flip onto the decoy. Grok no-ARM opaque is UNKNOWN 23/32 cyclic (decoy 3) and UNKNOWN 26/32 constant (decoy 5). Among those 15 commits, gold is 7 (Wilson [0.25, 0.70]), consistent with chance. OpenAI no-ARM cyclic at 4096/medium is a high gold rate (Table 4), not a chance-level commit rate; English names remain, so that gold cannot be attributed to a join over CONTEXT. Grok still prefers UNKNOWN over the decoy with no ARM line, so that preference is not a named-arm-only fact, and 7/15 does not test it.

Composer/Grok hashed-id H5-cyclic cells used `--preamble none` and score 6/32 and 0/32; the named-arm locks used the Cursor preamble and score 10/32 and 1/32. Both families go *down* with hashed ids, a sign reversal relative to OpenAI. We did not re-query Composer named-arm with the prompt file verbatim, so that reversal mixes case-id wording with preamble deletion and is not interpretable. Grok is at floor on both opaque two-path cells.

### 5.3 Both-opaque and readable relations

Table 5 is a second isolation suite on the same 32 records (`ANSWER_SEALED`; start seal in  $q$ ). Unique-path both-opaque is 32/32 for all three families. Composer/Grok relation-only unique (27/32, 30/32 in Table 7) is a different file, not evidence that hiding entities helps. Two-path both-opaque is OpenAI 1/32 (Wilson [0.01, 0.16]), Composer 0/32, Grok 0/32. A written hop recovers (32/32, 28/32, 32/32). OpenAI OE two-path (opaque entities, English relations) is 32/32: readable predicates restore end-to-end accuracy with anonymous nodes under this construction. Grok OE two-path remains 3/32 (UNKNOWN 29, decoy 0). On the  $n=200$  freeze (OpenAI), unique both-opaque and OE two-path are 200/200; both-opaque two-path is 12/200.

### 5.4 WikiMovies: hashed verbs and two-hop ARM

Table 6 uses the same actor→movie→director vs. writer template as MOVIE-A32 (person-valued tails). MOVIE-A32 unique-path with English verbs in  $q$  is 32/32 for all three families; opaque two-path is 17/32, 17/32, 0/32, with plan 32/32. A32’s ARM names two 2-hops; A100’s is `OPAQUE_AMBIG`  $n=100$ . On 32 A32 items matched by gold name (31 unique names) OpenAI is 17/32 vs. 29/32 under A100’s thinner ARM. A100 opaque two-path is 96/100 overall. A100 items 32–99 are 67/68 (index tail). That tail is not the name-complement of the 32 matched rows: A32 maps onto A100 indices 0–30 plus 41 (Gene Tierney is duplicated in A32), so index 41 sits in both counts and must not be added to 29/32. Keys differ; they

Table 5: WIKI-OO and opaque-entity (OE) on the same 32 graphs (start in  $q$ ). OpenAI OE two-path 32/32; both-opaque two-path 1/32. Grok OE two-path 3/32 (UNKNOWN 29). Plan is a control.

| Arm              | Ent.   | Rel.   | OpenAI | Composer | Grok  |
|------------------|--------|--------|--------|----------|-------|
| Unique path      | opaque | Eng.   | 32/32  | 32/32    | 32/32 |
| Two paths        | opaque | Eng.   | 32/32  | 28/32    | 3/32  |
| Unique path      | opaque | opaque | 32/32  | 32/32    | 32/32 |
| Two paths        | opaque | opaque | 1/32   | 0/32     | 0/32  |
| Two paths + plan | opaque | opaque | 32/32  | 28/32    | 32/32 |

Table 6: MOVIE-A100/MOVIE-B: same actor→movie→director vs. writer template as MOVIE-A32 (Miller et al., 2016; Zhang et al., 2018) (OpenAI  $n=100$ , start in  $q$ ). Condition B hashes verbs *and* uses a two-hop ARM: unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; coverage 13/100, selective accuracy 3/13). An archived ARM banner is 25/100.

| WikiMovies people (OpenAI, $n=100$ )                      | Unique  | Two-path |
|-----------------------------------------------------------|---------|----------|
| English relations, Condition A (verb in $q$ )             | 100/100 | 100/100  |
| Opaque relations, Condition A (verb in $q$ )              | 100/100 | 96/100   |
| Opaque relations, Condition B (verbs hashed; two-hop ARM) | 100/100 | 3/100    |
| Opaque two-path + plan, Condition B                       | —       | 100/100  |

cannot explain an English-name lookup. Director is always the first person-tail, so parametric knowledge and a listing-slot heuristic are not separated. Condition A leaves *directed/starred* in  $q$ ; we do not treat it as a selection result. Condition B hashes those verbs: OpenAI unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; Wilson [0.01, 0.08]). Coverage is 13/100 and selective accuracy 3/13. A written hop is 100/100. An archived Movie-B ARM banner is 25/100; the locked topology-only Condition B is 3/100. Composer/Grok Condition A two-path: 62/100, 36/100; Condition B  $n=100$  is OpenAI-only. On the  $n=200$  freeze, Condition A two-path is 122/200 (Wiki-H5 topology ARM, not A100’s ARM OPAQUE\_ambiguous  $n=100$  banner, so not a resample of 96/100); Condition B is 7/200. Movie-B’s ARM names two 2-hops (Wiki-H5-style); A100 Condition A does not. The contrast is two factors on WikiMovies (hashed verbs plus a two-hop ARM), not a same-prompt Wiki-H5 replication (Section 5.2). The  $n=32$  MetaQA-style template cell (Appendix H) is not MOVIE-B.

## 5.5 In-prompt seal reuse

Dual-path holds  $G$  fixed with two complete 2-hop routes and varies whether demonstrations expose the gold relation seals. Matched-relation is in-prompt reuse of shown correspondences, not induction of a general cipher. Novel-relation demos teach seals that do not appear on the gold route. An explicit join plan is the execution upper bound. Asymmetric vs. balanced noise: Appendix D.

Isolation  $n=32$  (Figure 2, Appendix D): OpenAI saturates matched-relation under both noise regimes (32/32, decoy 0). Novel-relation balanced is exact match 0/32: all 32 replies are stored UNKNOWN (24 empty completions, 8 explicit UNKNOWN). The novel arm’s expected gold is near zero. Novel-relation asymmetric is also 0/32 (UNKNOWN 24, of which 17 empty, decoy 8). An explicit dual-path plan scores 32/32. WikiMovies director/writer Dual is 32/32 vs. 0/32 for all three families (Composer novel decoy 2/32), with plan 32/32. On the synthetic Dual suite, Composer isolation-tag novel-relation UNKNOWN is a sealed-demo helper; Composer free-form is decoy 32/32. Grok free-form novel-relation asymmetric is UNKNOWN on all 32 items. Last-demo answer-token copy is 0/32 on every arm. On the  $n=200$  freeze (OpenAI), matched is 200/200 and novel is 2/200 (UNKNOWN 198 explicit, decoy 0), with plan 200/200.

On a two-path graph a forced uniform guess among the two complete routes is 1/2; that null does not apply once UNKNOWN is a third action. The pattern is consistent with prompt-local reuse of demonstrated seals and provides no evidence of systematic transfer to unseen relation tokens. We do not interpret this as a variable-binding mechanism. Arm definitions: Appendix D.

## 5.6 Missing-start control

The CEO isolation three-arm is an execution upper bound plus a missing-start cell, not the isolation of selection. Join plans score 32/32 on both plaintext and sealed atoms; SealRouter given the gold sealed relation sequence is 32/32; free sealed English scores 0/32 (all UNKNOWN) for Composer 2.5, OpenAI, and Grok 4.5 (Table 11). On the  $n=200$  freeze, plan remains 200/200 and free sealed English 0/200. We exclude a batched free-sealed cell that can succeed by returning a unique neighbor without using  $q$ .

## 6 Discussion

Sealed unique routes are executable (Table 3; hashed-id no-ARM unique 200/200 at 512 tokens). Under two competing sealed routes, named-arm OpenAI usually abstains (Table 3). Hashed case ids on the same graphs raise coverage and selective accuracy there; dropping the ARM line at 4096/medium is Table 4 (McNemar on the cyclic English/opaque pair). Unique no-ARM at 512 is not a matched-budget topology contrast with two-path no-ARM at 4096/medium. Alias  $k=3$  (Table 3) does not make the two 512-token cells meet. That surface move is large; it does not make abstention the wrong description of the *named-arm* cell. Grok no-ARM opaque still prefers UNKNOWN over the decoy (23 vs. 3 cyclic); among 15 commits gold is 7 (Wilson [0.25, 0.70]). OpenAI’s higher no-ARM gold rate is not a CONTEXT join (Section 5.2). English predicates restore OpenAI with anonymous nodes (32/32 vs. both-opaque 1/32); Grok OE two-path is 3/32 (UNKNOWN 29). Dual-path reuses shown seals (matched 200/200); the novel-seal arm is a negative control (2/200; expected gold near zero). A high English two-path rate does not imply the same join once relations are opaque. OIR withholds a retrieved-path ranking and a local plaintext decoder on the free arms; PrivGemo’s CWQ/WebQSP numbers keep those paths and answer locally (Tan et al., 2026). We do not claim mathematical unidentifiability. Hypothesis status: Table 10.

## 7 Limitations

OpenAI is the primary family. Headline  $n=200$  is the 15 September named-arm lock and the 512-token hashed-id lock; later alias draws do not replace them (Table 3; Section 5.1).  $n=32$   $k=3$  triples are Section 5.1. Composer/Grok stay at  $n=32$  on a different harness (Table 7). The competing two-path route is injected, not a Wikidata fact; graphs are four triples, smaller than retrieved KG-RAG subgraphs. The  $n=26$  decoy-0 slice is underpowered for a  $\sim 5\%$  decoy rate (Section 5.1). The  $n=32$  format token makes six P159 golds ill-posed;  $n=200$  does not use those golds. The  $n=32$  suite is 32 of 73 records, one template, one seed; decoys are cyclic, so item-independence for Wilson intervals is slightly optimistic (McNemar uses paired items but still assumes independent pairs).  $k=3$  at  $n=32$  mixes the August lock with later alias draws (Section 5.1). The two later OpenAI named-arm draws are both 3/32 and Grok both 0/32; Composer is 7/32 then 6/32; hashed-id is 21/32 then 22/32. August named-arm is the highest draw in all three families. This is not a generation-variance CI. English entities leave parametric cues. Named-arm opaque two-path keeps those names and is still mostly UNKNOWN (24/32, 151/200), not a dump of the headquarters city. We cannot separate a recalled HQ treated as inadmissible from a verb that does not map onto one seal. 2Wiki-CF Wikipedia leak is 75/200 and OpenAI overlap leak 50/200 (Appendix G). Both-opaque unique is 200/200 with names sealed. HMAC two-path rates vary by generator (6/32, 15/32, 12/32). Those runs are undated; keyed HMAC and random permutation both emit E+12hex tokens, so the 6 vs. 15 gap is not a token-shape contrast (Table 8). The paper’s central distinction between selection and traversal is inferred from the answer node rather than measured: we never score an emitted  $(r_1, r_2)$ . Named-arm OpenAI selective accuracy stays high; Grok no-ARM commits are 7 gold / 15 (Wilson [0.25, 0.70]; Section 5.2). Header $\times$ decoy at  $n=200$  is Table 4; Table 9 is the  $n=32$  header $\times$ decoy cell. Two-path K2 and no-ARM were not run at 512. MOVIE-A32 vs. MOVIE-A100 (17/32 vs. 29/32) is a thinner ARM on WikiMovies;  $n=200$  Condition A uses the Wiki-H5 topology ARM, not that banner. Dual novel OpenAI UNKNOWN includes empty completions. Forced-choice, scored  $(r_1, r_2)$ , and an HQ-city swap are not in this evaluation. The ARM line was not randomized independently of case-id wording on the original lock. Composer hashed-id H5-cyclic used `--preamble none`; the named-arm Composer lock used the Cursor preamble, so that sign reversal is uninterpretable.

## Broader Impact Statement

Public Wikidata CEO records (CC0; Vrandečić and Krötzsch, 2014) and the WikiMovies/MetaQA graph as released (Miller et al., 2016; Zhang et al., 2018). Appendix cells use 2WikiMultihopQA (Apache-2.0; Ho et al., 2020), Spider (CC-BY-SA-4.0; Yu et al., 2018), and FinQA as released (Chen et al., 2021). No personally sensitive information. Replies are hash-pinned, not a training dump. OIR is a diagnostic, not a confidentiality mechanism (English names remain on the primary  $2 \times 2$ ).

## 8 Conclusion

Unique-route sealed execution succeeds with the start in  $q$  (Table 3; hashed-id no-ARM unique 200/200 at 512 tokens; both-opaque unique 32/32 on three families, Composer/Grok on the Cursor SDK). On competing sealed routes, named-arm OpenAI usually abstains; hashed case ids raise exact-match on the same graphs (Table 3;  $k=3$  gaps 84–96). Exact-match under two sealed routes is strongly sensitive to prompt surface: case-id wording at 512 tokens; H5 vs. no-ARM McNemar at 4096/medium is Table 4 (cyclic and constant). Grok no-ARM opaque still prefers UNKNOWN over the decoy (23 vs. 3 cyclic); among 15 commits gold is 7 (Wilson [0.25, 0.70]; Section 5.2).

## References

- Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. FinQA: A Dataset of Numerical Reasoning over Financial Data. *EMNLP*, 2021.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. Faith and Fate: Limits of Transformers on Compositionality. *NeurIPS*, 2023.
- Ran El-Yaniv and Yair Wiener. On the Foundations of Noise-free Selective Classification. *JMLR*, 11:1605–1641, 2010.
- Zhouxiang Fang, Aayush Mishra, Muhan Gao, Anqi Liu, and Daniel Khashabi. ICL CIPHERS: Quantifying “Learning” in In-Context Learning via Substitution Ciphers. *EMNLP*, 2025.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. Talk like a Graph: Encoding Graphs for Large Language Models. *ICLR*, 2024.
- Jiahai Feng and Jacob Steinhardt. How do Language Models Bind Entities in Context? *ICLR*, 2024.
- Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks. *NeurIPS*, 2017.
- Shiyuan Guo, Henry Sleight, and Fabien Roger. All Code, No Thought: Current Language Models Struggle to Reason in Ciphered Language. *ICLR*, 2026.
- Tianyu Guo, Hanlin Zhu, Ruiqi Zhang, Jiantao Jiao, Song Mei, Michael I. Jordan, and Stuart Russell. How Do LLMs Perform Two-Hop Reasoning in Context? *arXiv:2502.13913*, 2025.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing A Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps (2WikiMultihopQA). *COLING*, 2020.
- H. Krawczyk, M. Bellare, and R. Canetti. HMAC: Keyed-Hashing for Message Authentication. IETF RFC 2104, 1997.
- Chenhui Liu, Jianpeng Zhou, and Jiahai Wang. Chain-of-Relations: Faithful and Efficient LLM Reasoning over Knowledge Graphs via Relation-Centric Exploration. *Findings of ACL*, 2026. <https://aclanthology.org/2026.findings-acl.2138/>.

- Linhao Luo, Yuan-Fang Li, Gholamreza Haffari, and Shirui Pan. Reasoning on Graphs: Faithful and Interpretable Large Language Model Reasoning. *ICLR*, 2024.
- Quinn McNemar. Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages. *Psychometrika*, 12(2):153–157, 1947.
- Alexander Miller, Adam Fisch, Jesse Dodge, Amir-Hossein Karimi, Antoine Bordes, and Jason Weston. Key-Value Memory Networks for Directly Reading Documents (WikiMovies). *EMNLP*, 2016.
- Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncl Tuzel, Samy Bengio, and Mehrdad Farajtabar. GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models. *ICLR*, 2025.
- Yunfeng Ning, Mayi Xu, Jintao Wen, Qiankun Pi, Yuanyuan Zhu, Ming Zhong, Jiawei Jiang, and Tiejun Qian. Privacy-protected Retrieval-Augmented Generation for Knowledge Graph Question Answering (ARoG). *AAAI*, 2026.
- Stephen Robertson and Hugo Zaragoza. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.
- Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying Language Models’ Sensitivity to Spurious Features in Prompt Design. *ICLR*, 2024.
- Jiashuo Sun, Chengjin Xu, Luminyuan Tang, Saizhuo Wang, Chen Lin, Yeyun Gong, Lionel Ni, Heung-Yeung Shum, and Jian Guo. Think-on-Graph: Deep and Responsible Reasoning of Large Language Model on Knowledge Graph. *ICLR*, 2024.
- Xingyu Tan, Xiaoyang Wang, Qing Liu, Xiwei Xu, Xin Yuan, Liming Zhu, and Wenjie Zhang. PrivGemo: Privacy-Preserving Dual-Tower Graph Retrieval for Empowering LLM Reasoning with Memory Augmentation. arXiv:2601.08739, 2026.
- Alon Talmor and Jonathan Berant. The Web as a Knowledge-Base for Answering Complex Questions. *NAACL*, 2018.
- Xiaojuan Tang, Zilong Zheng, Jiaqi Li, Fanxu Meng, Song-Chun Zhu, Yitao Liang, and Muhan Zhang. Large Language Models are In-Context Semantic Reasoners rather than Symbolic Reasoners. arXiv:2305.14825, 2023.
- Denny Vrandečić and Markus Krötzsch. Wikidata: A Free Collaborative Knowledgebase. *CACM*, 2014.
- Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. Can Language Models Solve Graph Problems in Natural Language? *NeurIPS*, 2023.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *NeurIPS*, 2022.
- Jerry Wei, Le Hou, Andrew Lampinen, Xiangning Chen, Da Huang, Yi Tay, Xinyun Chen, Yifeng Lu, Denny Zhou, Tengyu Ma, and Quoc Le. Symbol tuning improves in-context learning in language models. *EMNLP*, 2023.
- Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. Know Your Limits: A Survey of Abstention in Large Language Models. *TACL*, 13:529–556, 2025.
- Edwin B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. *JASA*, 22(158):209–212, 1927.
- Wen-tau Yih, Matthew Richardson, Chris Meek, Ming-Wei Chang, and Jina Suh. The Value of Semantic Parse Labeling for Knowledge Base Question Answering (WebQSP). *ACL*, 2016.

Tao Yu, Rui Zhang, Kai Yang, Michihiro Yasunaga, Dongxu Wang, Zifan Li, James Ma, Irene Li, Qingning Yao, Shanelle Roman, Zilin Zhang, and Dragomir Radev. Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL. *EMNLP*, 2018.

Yuyu Zhang, Hanjun Dai, Zornitsa Kozareva, Alexander J. Smola, and Le Song. Variational Reasoning for Question Answering with Knowledge Graph. *AAAI*, 2018.

## A Protocol, scoring, and models

Quizzes (`runs/`) and locks (`results/`): the supplementary archive (Python  $\geq 3.10$ ; hash-pin `results/SHA256SUMS`, not `HEAD`). OpenAI: `gpt-5.6-sol`, no tools, temperature omitted. Named-arm, 512-token hashed-id, and unique no-ARM sidecars store `returned_model gpt-5.6-sol` (vendor alias, not a dated snapshot pin). August  $n=32$  stores the parsed answer, not `model/usage`. Named-arm and 512-token hashed-id omit `reasoning_effort`. Header $\times$ decoy at 4096 sets `reasoning_effort` medium.  $k=3$  sidecars also store `returned_model`. Named-arm  $n=200$  OPAQUE\_AMBIG sidecars are `finish_reason stop` on 200/200; no-ARM cyclic OPQ\_NONE\_CYC likewise 200/200 `stop`; hashed-id 512 OPQ\_H5\_CYC is `stop` 199 and `length 1` (the `NO_OUTPUT` item). August  $n=32$  stores no usage. Composer/Grok: Cursor agent SDK and a reply-only preamble OpenAI never received. Cite Dual free-form as `*ff`. Empty Dual/no-hop completions were stored as `UNKNOWN`; Wiki-H5 and Wiki-OO were not. We never impute n.r. as zeros.

WIKI-H5 seed 20260821, 16-byte keys from `oir-fact-2x2-iso-v1`; 32 of 73 CEO rows; decoys  $i+1 \bmod 32$  (Wilson slightly optimistic). Wiki-H5  $n=32$  isolation is  $32 \times 5$  named arms;  $n=32$  header $\times$ decoy is 224 queries per family;  $n=200$  header $\times$ decoy at 4096/medium is 1400. Hashed-id unique-path no ARM (16 September) is 27/32 (26/26 city) and 200/200 (512 tokens; same `CONTEXT` as `OPAQUE_UNIQUE`). WIKI-OO uses `oir-entrel-2x2-iso-v1`; CEO-NL seed 20260820. The  $n=12/n=24$  three-arm is `batch` (one shared key `oir-ceiling-three-arm-v1`); isolation  $n=32$  is per-item keys. WIKI-H5-200 is a separate freeze (`data/real/wikidata_ceo_hops_n200.json`, seed 20260915;  $P31 \in \{Q515 \text{ city}, Q1549591 \text{ big city}, Q1093829 \text{ city of the United States}, Q1637706 \text{ city with millions of inhabitants}, Q3957 \text{ town}, Q5119 \text{ capital}, Q13539802 \text{ place with town rights and privileges}, Q5153359 \text{ municipality of the Czech Republic}, Q7930989 \text{ city or town}\}$ ; SPARQL verify 200/200). Labels are Wikidata English labels: Vienna/The Hague/Mattersburg use `Q13539802`; Prague/Čáslav use `Q5153359`. HMAC `oir-fact-2x2-iso-n200-v1`;  $n=32$  prompts are not rewritten. WikiMovies  $n=200$  keeps the  $n=100$  prefix and adds 100 actors.

**Run-tag manifest.** Body text names cells by packing and date. Tags:

| Cell                                            | Tag                                                       |
|-------------------------------------------------|-----------------------------------------------------------|
| Named-arm $n=32$ August                         | <code>gpt56</code>                                        |
| Named-arm $n=32$ 15 Sep                         | <code>gpt56k3r2/r3</code>                                 |
| Named-arm $n=32$ 11 Sep 512 (OPAQUE_AMBIG only) | <code>gpt56sep</code>                                     |
| Hashed-id $n=32$ 11 Sep 512                     | <code>gpt56abl512</code>                                  |
| Hashed-id $n=32$ 4096/medium                    | <code>gpt56abl</code>                                     |
| Named-arm $n=200$ 15 Sep                        | <code>gpt56n200</code>                                    |
| Hashed-id $n=200$ 512                           | <code>gpt56n200h512</code>                                |
| Header $\times$ decoy $n=200$ 4096/medium       | <code>gpt56n200abl</code>                                 |
| $n=200$ $k=3$ 16 Sep                            | <code>*k3r2/*k3r3</code> on those two 512 stems           |
| Unique no-ARM $n=200$ / $n=32$ 512              | <code>gpt56n200uninone</code> / <code>gpt56uninone</code> |
| $n=32$ shuffle                                  | <code>gpt56</code> on the shuffle stem                    |
| Grok named-arm English verbatim                 | <code>grok45pn</code>                                     |
| Composer Dual isolation / free-form             | <code>composer25</code> / <code>composer25ff</code>       |
| Grok Dual free-form                             | <code>grok45ff</code>                                     |

Lock JSON lives under `results/` with that tag as the filename suffix (e.g. `...gpt56n200.json`, `...gpt56n200h512.json`). `gpt56n200h512` queried only `OPQ_H5_CYC`. The  $n=200$  shuffle stem is `factorial_2x2_iso_shuffle_n200_` (case ids `F200S_` rather than `F200_`). It is not a second model snapshot and does not overwrite the headline lock.

Non-city P159 golds: **Broadway**, **NRK\_Marienlyst**, **Fox\_Plaza**, **CNN\_Center**, **Lod\_Railway\_Station**; **New\_Jersey** is a state. OpenAI two-path decoys are items 27 and 31 (both non-city). City slice 5/26 (decoy 0) is underpowered;  $n=200$  is decoy 11. An archived Movie-B ARM banner is 25/100; locked topology-only B is 3/100. CEO free English hashes  $q$  piecewise.

**Normalization and HMAC.** Strip; collapse whitespace to `_`; map other non-alphanumeric characters (except underscore, period, hyphen) to `_` using `[A-Za-z0-9_.-]` (Zürich→Z\_rich; pure CJK keeps Unicode); truncate to 64; empty maps to **EMPTY**. The CEO JSON already deleted diacritics upstream. Distinct raw atoms must normalize to distinct strings. HMAC-SHA256 is over UTF-8 bytes with a per-item 16-byte key; the token is **E** plus the first 6 digest bytes as 12 hex digits. Truncation is not injective over  $\{0,1\}^*$ ; 48-bit width is instance-wise injective on these small atom sets. Entity and relation types are not domain-separated. On each scored Wikidata  $2 \times 2$  instance the normalized entity strings and relation strings are disjoint, and no HMAC collision among prompt atoms was observed. Sealing is deterministic given the instance key; the same plaintext need not map to the same token across prompts.

**Composer/Grok named-arm ( $n=32$ ).** Table 7 is the SDK roster, not a ranking of families. **UNKNOWN** is an explicit line (missing/unparsed 0; both-named 0). Composer “other” is truncated gold Lod on **ENG\_UNIQUE**, **OPAQUE\_UNIQUE**, and plan, plus Los Angeles for gold **Fox\_Plaza** on **OPAQUE\_UNIQUE**.

Table 7: Four-way outcomes: gold / decoy / **UNKNOWN** / other. Composer 2.5 and Grok 4.5 used Cursor’s agent SDK with a reply-only preamble OpenAI never received. Body Table 3 is the OpenAI lock.

| Wikidata named-arm (iso. $n=32$ ) | OpenAI   | Composer 2.5 | Grok 4.5 |
|-----------------------------------|----------|--------------|----------|
| English, unique                   | 31/0/1/0 | 29/0/2/1     | 32/0/0/0 |
| English, two paths                | 31/0/1/0 | 26/0/6/0     | 4/0/28/0 |
| Opaque, unique                    | 32/0/0/0 | 27/0/3/2     | 30/0/2/0 |
| Opaque, two paths                 | 6/2/24/0 | 10/0/22/0    | 1/0/31/0 |
| Opaque two paths + plan           | 32/0/0/0 | 28/0/3/1     | 32/0/0/0 |

## B Generator robustness and $n=32$ header×decoy

Table 8: Generator robustness on the same 32 Wikidata graphs and question template. Unique-path accuracy exceeds two-path accuracy under three independently generated opaque relation maps; the two-path *rate* is not invariant. Wilson:  $29/32 \Rightarrow [0.76, 0.97]$ ,  $12/32 \Rightarrow [0.23, 0.55]$ ,  $15/32 \Rightarrow [0.31, 0.64]$ . Alphanumeric two-path is all abstention or gold (decoy 0).

| Opaque generator (OpenAI, start in $q$ ) | Unique | Two-path |
|------------------------------------------|--------|----------|
| Keyed HMAC ( <b>E</b> +12hex)            | 32/32  | 6/32     |
| Random permutation ( <b>E</b> +12hex)    | 31/32  | 15/32    |
| Random alphanumeric (not HMAC-shaped)    | 29/32  | 12/32    |

A random permutation of HMAC-shaped labels on the same graphs preserves unique > two-path (OpenAI 31/32 vs. 15/32; Composer 28/32 vs. 2/32; Grok 31/32 vs. 0/32). An English-entity ambiguity sweep  $k=1..5$  (constant decoy) keeps OpenAI at 29, 29, 28, 27, 28; Composer falls to 11/32; Grok is 0/32 for  $k \geq 2$ . Disney names remain, so  $k$  is not isolated from entity semantics.

Header×decoy uses hashed case ids that do not name the condition on the same 32 people (**header\_decoy\_ablation\_iso\_\*.json**; body Section 5.2). Cells are named by header and decoy: H5 keeps the Wiki-H5 ARM line, K2 keeps a shorter two-hop ARM, and “no ARM” deletes the line; cyclic decoys attach record  $i+1$ , constant decoys reuse the first pool record that does not share company, HQ, or person ( $n=32$ : Cogent Communications → Washington, DC, gold=decoy 0;  $n=200$ : San\_Jose  $\times 199$ , London  $\times 1$ , gold=decoy 0). OpenAI 4096/medium  $n=32$  golds:

Table 9: Header×decoy gold counts on the same 32 people. English, no ARM, cyclic is 29/32. H5 McNemar 14 vs. 4,  $p=0.031$ ; no-ARM gold 25 vs. 27, McNemar 1 vs. 3,  $p=0.625$ . Body Table 4 is the  $n=200$  lock.

| ARM line (OpenAI, hashed ids, 4096/medium) | Cyclic | Constant |
|--------------------------------------------|--------|----------|
| no ARM                                     | 25/32  | 27/32    |
| H5                                         | 20/32  | 10/32    |
| K2                                         | 30/32  | 26/32    |

Composer/Grok the same day, prompt file verbatim (`--preamble none`): H5 6/2 and 0/0; K2 20/13 and 0/0; no ARM 21/18 and 6/1; English 26 and 24. Named-arm English Grok verbatim (`grok45pn`) stays 4/32: the Cursor preamble is not that cell. `ENG_NONE_CYC` 24/32 differs in both ARM and case ids (Grok 24/32 vs. no-ARM opaque 6/32; McNemar 18 vs. 0). Grok no-ARM opaque is cyclic 6/3/23 (gold / decoy / UNKNOWN) and constant 1/5/26; among 15 commits, gold is 7 (Wilson [0.25, 0.70]). Constant decoy does not raise the Wiki-H5-header cell on any family.

## C Hypothesis status

Table 10: Hypothesis status. H4 exact-match is OpenAI Dual; Composer free-form novel is decoy 32/32 (`composer25ff`). H5 is named-arm WIKI-H5 (OpenAI). H6 hashes verbs *and* uses a two-hop ARM; an archived banner is 25/100. H7 is OpenAI-specific (OE vs. WIKI-OO). H8 is prompt surface on the same graphs (case-id +96 is Table 3; ARM-line McNemar is Table 4), not cyclic vs. constant (which changes CONTEXT).

| Hypothesis                                                                 | Test                                                   | Status                                                                                                                                            |
|----------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| H1. Opacity alone blocks traversal                                         | unique-path, start in $q$                              | not supported (32/32)                                                                                                                             |
| H2. Written hop restores e2e (control)                                     | two-path + plan; SealRouter                            | control (32/32)                                                                                                                                   |
| H3. Input-binding control (CEO piecewise $q$ )                             | CEO isolation NL                                       | control (0/32); not selection                                                                                                                     |
| H4. Matched demos restore shown seals                                      | dual-path matched vs. novel                            | OpenAI 32/32 vs. 0/32                                                                                                                             |
| H5. End-to-end accuracy degrades when opaque relations meet path ambiguity | matched 2×2, start in $q$                              | named-arm $n=32$ McNemar 25 vs. 0 (city 21 vs. 0); $n=200$ no-ARM hashed cyclic English vs. opaque McNemar 25 vs. 2 (4096/medium); Secs. 5.1, 5.2 |
| H6. Movie KB, hashed verbs + two-hop ARM (two-factor check)                | WikiMovies Cond. B $n=100$                             | OpenAI two-path 3/100; archived banner 25/100                                                                                                     |
| H7. Readable relations with opaque entities                                | Wikidata OE vs. OO                                     | supported for OpenAI here                                                                                                                         |
| H8. Prompt surface moves two-path exact match on the same graphs           | case-id at 512 cyclic; ARM at 4096 cyclic and constant | $n=200$ case-id +96; H5 vs. no-ARM McNemar cyclic 6 vs. 51, constant 4 vs. 111; Sec. 5.2                                                          |

Hypothesis H5 in this table reports the named-arm WIKI-H5 McNemar and the no-ARM hashed cyclic contrast; H5 in Section 5.2 is the ARM header.

Table 11 is the missing-start CEO-NL control, not WIKI-H5.

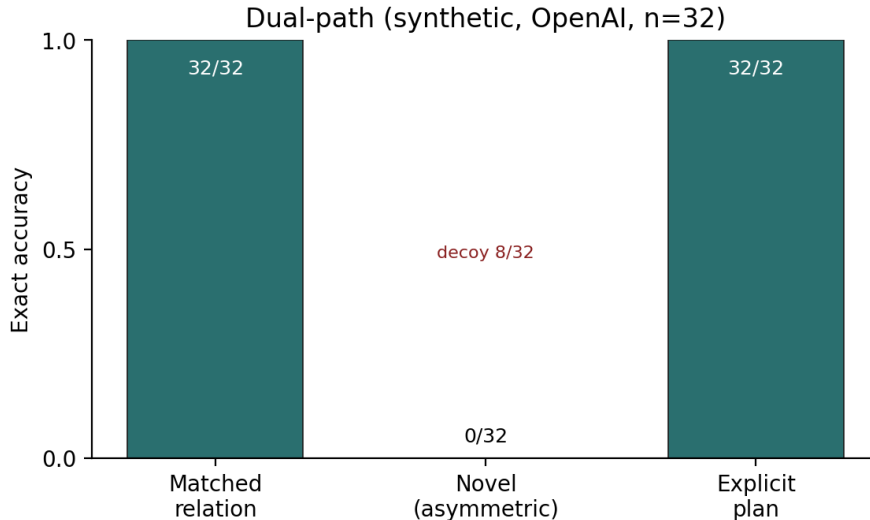


Table 11: Missing-start control, not WIKI-H5. Isolation  $n=32$ . Batch  $n=12/n=24$  uses one shared key (Appendix A).

| CEO-NL isolation $n=32$              | Composer 2.5 | OpenAI | Grok 4.5 |
|--------------------------------------|--------------|--------|----------|
| Sealed plan / SealRouter             | 32/32        | 32/32  | 32/32    |
| Free sealed English (piecewise $q$ ) | 0/32         | 0/32   | 0/32     |

## D Dual-path arms

We name the arms by what the demos teach. **Asymmetric noise:** extra edges hang only off the gold midpoint (the trap path looks cleaner). **Balanced noise:** symmetric extra edges on both midpoints. **Matched-relation** (asymmetric or balanced): demos expose the gold relation seals; the decoy path uses different seals. **Novel-relation** (asymmetric or balanced): demos teach seals that do not appear on the gold route; both routes are novel at quiz time. **Explicit join plan:** gold relation sequence is written in the prompt.

Figure 2: DUAL (synthetic, OpenAI  $n=32$  isolation): matched 32/32 under both noise regimes (one bar; both 32/32); novel asymmetric 0/32 (decoy 8); explicit plan 32/32.

Isolation  $n=32$  is the reported cell. Cite Dual free-form as `composer25ff/grok45ff`; the isolation-tag Composer helper is not free-form. Local `mistral/qwen35` are not in the roster (plan n.r.).

## E No-hop binding (JSON)

Prose labels in Table 12 map to lock JSON arms: schema list `LEGEND` and English protocol `META` (`seal_layer_legend_small_n32_harness*.json`); silent `ISO_SILENT`, equality recipe `ISO_RULE`, and broken match `NS_SILENT` (`opaque_iso_json_n32_harness*.json`); two-route recipe `TWO_REL` and vague hints `ABSTRACT` (`opaque_hint_reason_n32_harness*.json`). Prompts are fixed English blocks, not WIKI-H5. Equality means token identity (Appendix J). English JSON keys alone do not restore the walk (1/6 on the  $n=6$  batch; that arm is not run at  $n=32$ ).

Table 12: Selection support without a written hop (Wikidata CEO JSON; city or mentor country is the distractor; isolation; per-item HMAC). Schema-only and broken token overlap fail for all three. An English protocol or an equality recipe that does not name the field saturates. Silent same- $\sigma$  overlap and vague hints are family-dependent, not a three-family restore. OpenAI silent 6/32 and vague-hint 13/32 include 24 and 15 empty completions stored as UNKNOWN. \*Composer vague-hint distractor 1/32. JOIN was not evaluated at  $n=32$ . Wilson: 32/32  $\Rightarrow$  [0.89, 1.00], 0/32  $\Rightarrow$  [0.00, 0.11], 6/32  $\Rightarrow$  [0.09, 0.35], 31/32  $\Rightarrow$  [0.84, 0.99]. Silent same- $\sigma$  OpenAI 6/32 is a JSON-binding cell, distinct from WIKI-H5. NS\_SILENT ARM lines say “Match impossible”; those 0/32 rows are not a sealed-join collapse.

| Arm (no written hop)               | OpenAI | Composer 2.5 | Grok 4.5 |
|------------------------------------|--------|--------------|----------|
| Schema list only                   | 0/32   | 0/32         | 0/32     |
| English protocol (no hop)          | 32/32  | 32/32        | 32/32    |
| Same $\sigma$ , silent             | 6/32   | 19/32        | 31/32    |
| Same $\sigma$ + equality recipe    | 32/32  | 32/32        | 32/32    |
| JSONKEY: match broken              | 0/32   | 0/32         | 0/32     |
| Sealed Q, two routes, recipe       | 31/32  | 25/32        | 32/32    |
| Sealed Q, two routes, vague hints* | 13/32  | 23/32        | 31/32    |

## F Sealing middleware

Bind a typed full name to one graph atom, then seal. Isolation without binding is 0/6 vs. layer 6/6 (Composer, OpenAI, unversioned Claude; plaintext in the packed prompt: 0). Those files attach a join: an execution control, not the free WIKI-H5 6/32 cell.

## G Packaging probes

Not the matched  $2 \times 2$ . Free sealed English 0/12 vs. plan 12/12 on table/JSON/vault (Composer, OpenAI). Spider identifier-in- $q$   $n=100$ : EQ 51/100, namespace-break 0/100 (OpenAI; not official EM). If the identifier is not in  $q$  ( $n=26$ ), the namespace-break cell is 20/26: an English  $q$  plus legend can still bind. Excel NL names both predicates: Composer 0/16 vs. plan 16/16. FinQA NL 0/8 vs. gold programs 8/8 (Chen et al., 2021). 2Wiki-CF: Composer plain NL 200/200; OpenAI 150/200 with overlap leak 50/200 (not a 2Wiki eval). Long context: NL 0/8, plans 8/8. Rename under a second HMAC: 12/12. SealRouter and gold-SQL are non-LLM controls (Figure 3): gold-SQL is the Spider gold query executed after name sealing (191/200), not an exact join.

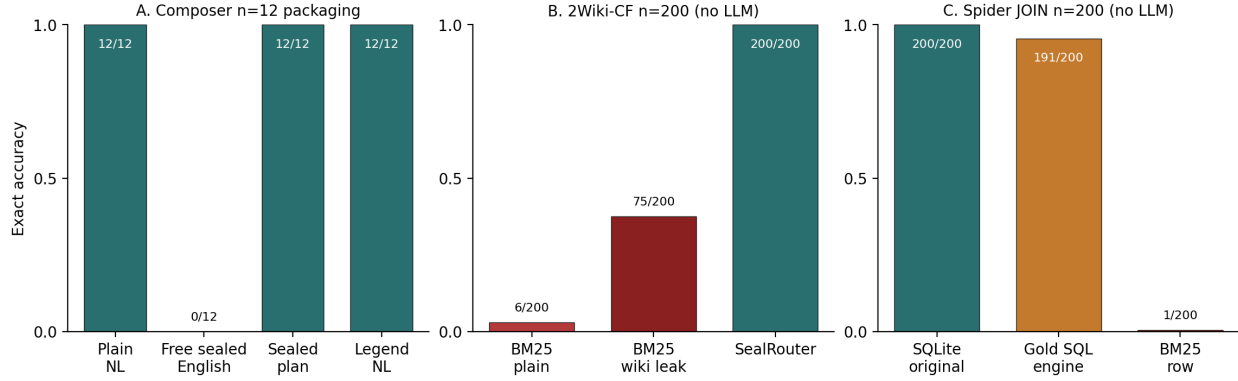


Figure 3: Non-LLM and packaging controls. Left: Composer  $n=12$  (plain 12/12, free sealed English 0/12, plan 12/12, legend 12/12). Middle: 2Wiki-CF lexical BM25 6/200 (not a 2Wiki eval), Wikipedia leak 75/200, SealRouter 200/200. Right: Spider sqlite  $n=200$  (200/200), gold-SQL 191/200 (sealed names; 9 execution misses), lexical BM25 1/200 (Robertson and Zaragoza, 2009). Spider identifier-in- $q$  EQ 51/100 is a different cell.

## H Extended scores

MetaQA-style 2-hop templates on the same public movie graph are a question-form control, not an evaluation on the official MetaQA 14K test split. Plaintext is 32/32 for all three families. Condition A (English question over opaque relations) is OpenAI 32/32, Composer 19/32, Grok 15/32. Condition B hashes those verbs except [start]: OpenAI 25/32, Composer 27/32, Grok 14/32 (plans 32/32, 31/32, 32/32). That  $n=32$  template cell is not MOVIE-B.

## I Sign-flip of the two-path opacity contrast

Let  $d_i = \mathbf{1}[\text{opaque correct}] - \mathbf{1}[\text{English correct}]$  and  $c_i = d_i^{\text{two}} - d_i^{\text{unique}}$ . Of 32 WIKI-H5 items, 24 have  $c_i = -1$ , one has  $c_i = -2$ , 7 have  $c_i = 0$ ;  $T = -26$ . Unique-path opaque is 32/32, so  $d^{\text{unique}}$  is nonzero on one item and the  $c_i$  mostly copy the two-path McNemar. The construction is an item-level interaction contrast whose null is exchangeability of unique-path and two-path opacity effects ( $p = 2^{-24} = 5.96 \times 10^{-8}$ ). With unique-path at ceiling it cannot say anything beyond the two-path simple effect, so the identical  $p$  with McNemar 25 vs. 0 (McNemar, 1947) is expected. Opacity is not a randomized within-item treatment. The body reports Wilson intervals and the paired counts.

## J Equality recipe (JSON cell)

Verbatim protocol used in Table 12, *not* in WIKI-H5: “QUESTION and CONTEXT use the same encoding. Identical tokens are the same entity or the same field (equality, not English). START is a QUESTION token that also appears as an id value in CONTEXT. The asked field is a QUESTION token that also appears as a JSON key. Walk CONTEXT by token equality. If more than one reading is possible, UNKNOWN.”